

Full Canonical Regeneration Plan

Created: 2026-03-22

Status: PLANNING ONLY — Do NOT execute without explicit approval

Scope: `scripts/run_production_v2.py` , active analysis generators, `results/baseline/` , `results/analysis/` , `results/figures/`

=== 1. REGENERATION EXECUTION PLAN ===

Phase 1: Baseline Production (~2-3 minutes)

Step	Command	Outputs	Est. Time
1a	<code>python scripts/ run_production_v2.py --reps 30</code>	All <code>results/ baseline/</code> artifacts	~100s

Outputs regenerated (complete list):

- `results/baseline/allocations/allocations_K{10,15,20,25,30,35,40,45,48}.csv` (9 files)
- `results/baseline/simulation/results_K{10..48}.csv` (9 files)
- `results/baseline/simulation/all_results_raw.csv` (1 file — 810 rows: 3 policies × 9 K × 30 reps)
- `results/baseline/tables/` — 18 statistical tables (descriptive, ANOVA, posthoc, effect sizes, CIs, queue, exp pivots, publication tables 1-4 in CSV+TeX)
- `results/baseline/figures/` — 44 figures (experiment panels, allocation maps, RT distributions, fleet sensitivity, validation, V&V, pub_fig1-5)
- `results/baseline/comparison_with_v1.csv`
- `results/baseline/experiment_log.txt`

What this script does internally (6 steps):

1. Generate allocations for $P0/P1/P2 \times K=\{10..48\}$
2. Run 810 simulation replications (30 reps × 27 scenarios)
3. Perform ANOVA + Tukey HSD + effect sizes
4. Generate all baseline figures
5. Create v1 comparison table
6. Write experiment log

Prerequisites:

- `data/processed/` must be fully populated (distance matrix, demand lambda tables, firehouses, precincts)
- PuLP solver available for P2 MIP
- All `src/ems_readiness/` modules import clean

Phase 2: Verification & Validation (~15s)

Step	Command	Outputs	Est. Time
2a	<code>python scripts/run_verification.py</code>	<code>results/baseline/simulation/verification/0{1..4}_*.json</code>	~5s
2b	<code>python scripts/run_validation_pilots.py</code>	<code>results/baseline/simulation/validation_pilot/*.json + *.csv</code>	~10s

Outputs:

- Verification: `01_toy_example.json`, `02_zero_demand.json`, `03_single_unit.json`, `04_extreme_demand.json`
 - Validation: `pilot1_p0_vs_p2.json`, `pilot1_comparison_table.csv`, `pilot2_sensitivity_K.json`, `pilot2_sensitivity_K_table.csv`, `pilot3_sensitivity_demand.json`
-

Phase 3: Analysis Artifacts (~5-10 minutes)

Step	Script	Outputs	Est. Time	Data Source
3a	scripts/analysis/regenerate_all_figures.py	results/figures/ (21 canonical figures)	~30s	results/baseline/simulation/all_results_raw.csv (V2 data)
3b	scripts/analysis/analyze_production_results.py	results/analysis/tables/ (8 tables)	~10s	⚠ results/analysis/simulation/production/ (V1 data, CAP=5)
3c	scripts/analysis/generate_publication_figures.py	results/analysis/figures/pub_fig* (5 figs)	~15s	⚠ results/analysis/simulation/production/ (V1 data, CAP=5)
3d	scripts/analysis/generate_precinct_demand_visualizations.py	results/figures/precinct_demand_*.png (2 figs)	~10s	data/processed/
3e	scripts/analysis/analyze_queue_metrics.py	results/figures/queue_*.png , results/tables/queue_*.csv	~10s	⚠ V1-DATA refs
3f	scripts/analysis/analyze_seasonal_patterns.py	results/figures/seasonal_*.png , results/tables/seasonal_*.csv	~15s	data/processed/crashes_manhattan.csv
3g	scripts/analysis/generate_summary_dashboard.py	results/figures/project_summary_dashboard.png	~5s	⚠ V1-DATA refs, CAP=5
3h	scripts/analysis/capacity_sensitivity_full_spectrum.py	results/analysis/capacity_comparison / (full cap sweep)	~3 min	Runs own simulations

Step	Script	Outputs	Est. Time	Data Source
3i	scripts/analysis/generate_capacity_sensitivity_heatmap.py	results/figures/capacity_sensitivity_heatmap.png	~5s	results/analysis/capacity_comparison/
3j	scripts/analysis/generate_all_heatmaps.py	results/analysis/heatmaps/ (108+ maps)	~2 min	Runs own allocations
3k	scripts/analysis/run_cbd_experiment.py	results/analysis/simulation/cbd_experiment/	~30s	Runs own simulations
3l	scripts/analysis/run_cbd_focused_optimization.py	results/analysis/cbd_focused_comparison/	~30s	⚠ CAP=5
3m	scripts/analysis/run_distance_comparison_experiment.py	results/analysis/distance_comparison/	~30s	⚠ CAP=5
3n	scripts/analysis/p0_spatial_analysis.py	results/analysis/figures/p0_spatial*.png	~10s	data/processed/
3o	scripts/analysis/generate_tradeoff_improved.py	results/figures/response_time_coverage_tradeoff*.png	~5s	⚠ CAP=5

⚠ CRITICAL: V1-vs-V2 Data Source Conflict

Several analysis scripts (3b, 3c, 3e, 3g, 3l, 3m, 3o) still read from `results/analysis/simulation/production/` which contains **V1 data (cap=5)**. These scripts need to be either:

1. **Redirected** to read from `results/baseline/simulation/all_results_raw.csv` (V2, cap=2), OR
2. **Marked as V1-legacy** and excluded from canonical regeneration

Phase 4: Unit Tests (~30s)

Step	Command	Outputs	Est. Time
4a	<code>python -m pytest tests/ -v</code>	Console output (39 tests across 4 modules)	~30s

Phase 5: Documentation Reconciliation (~manual, 30-60 min)

After all artifacts are regenerated, update docs to reference correct files and metrics. See Section 3 below.

=== 2. CANONICAL ARTIFACT LIST ===

Baseline Simulation (Source of Truth)

Artifact	Path	Generated By
Raw simulation data (810 rows)	<code>results/baseline/simulation/all_results_raw.csv</code>	<code>run_production_v2.py</code>
Per-K simulation results (9 files)	<code>results/baseline/simulation/results_K{N}.csv</code>	<code>run_production_v2.py</code>
Verification JSONs (4 files)	<code>results/baseline/simulation/verification/0{1..4}_*.json</code>	<code>run_verification.py</code>
Validation pilot JSONs (3 files)	<code>results/baseline/simulation/validation_pilot/pilot{1..3}_*.json</code>	<code>run_validation_pilots.py</code>
Validation comparison tables (2 files)	<code>results/baseline/simulation/validation_pilot/pilot{1,2}_*_table.csv</code>	<code>run_validation_pilots.py</code>

Baseline Tables (18 files)

Artifact	Path
Descriptive statistics	results/baseline/tables/descriptive_statistics.csv
ANOVA results	results/baseline/tables/anova_results.csv
Post-hoc comparisons	results/baseline/tables/posthoc_comparisons.csv
Confidence intervals	results/baseline/tables/confidence_intervals.csv
Effect sizes	results/baseline/tables/effect_sizes.csv
Queue statistics	results/baseline/tables/queue_statistics.csv
Queue ANOVA	results/baseline/tables/queue_anova.csv
Sensitivity summary	results/baseline/tables/sensitivity_summary.csv
Exp1 summary	results/baseline/tables/exp1_summary.csv
Exp2–4 pivot tables	results/baseline/tables/exp{2,3,4}_pivot_rt.csv
Production results	results/baseline/tables/production_results.csv
Seasonal analysis	results/baseline/tables/seasonal_analysis.csv
Statistical analysis	results/baseline/tables/statistical_analysis.csv
Publication Tables 1–4	results/baseline/tables/table{1..4}_*.csv + .tex
Validation results	results/baseline/tables/validation_results.csv

Baseline Figures (44 files)

EDA/Demand (10): fig_hourly_demand.png , fig_daily_demand.png , fig_temporal_trends.png , fig_crash_heatmap.png , fig_precinct_density.png , fig_precinct_demand.png , fig_demand_model_fit.png , fig_hourly_rates.png , fig_cbd_comparison.png , fig_firehouses_map.png

Optimization (5): `fig_policy_comparison.png` , `fig_tradeoff_curve.png` , `policy_comparison.png` , `policy_comparison_panel_K20_cap2.png` , `p0_vs_p2_response_time.png`

Experiment panels (4): `exp1_policy_comparison.png` , `exp2_fleet_sensitivity.png` , `exp3_demand_sensitivity.png` , `exp4_service_robustness.png`

Fleet/capacity (7): `fleet_sensitivity_v2_dual.png` , `production_fleet_sensitivity.png` , `mean_rt_vs_K.png` , `p95_rt_vs_K.png` , `coverage_vs_K.png` , `utilization_vs_K.png` , `queue_metrics_vs_K.png`

V&V (4): `verification_toy_timeline.png` , `validation_p0_vs_p2.png` , `validation_sensitivity_K.png` , `validation_sensitivity_demand.png`

Publication (5): `pub_fig1_policy_comparison.png` , `pub_fig2_fleet_sensitivity.png` , `pub_fig3_demand_robustness.png` , `pub_fig4_service_sensitivity.png` , `pub_fig5_performance_heatmap.png`

Allocation maps (3): `allocation_map_K{20,30,40}.png`

RT distributions (3): `rt_distribution_K{20,30,40}.png`

Other (3): `response_time_distribution_by_policy.png` , `statistical_effect_sizes.png` , `effect_sizes.png` , `project_summary_dashboard.png`

Baseline Allocations (9 files)

`results/baseline/allocations/allocations_K{10,15,20,25,30,35,40,45,48}.csv`

Canonical Top-Level Figures (`results/figures/` , 21 files)

These are generated by `regenerate_all_figures.py` and individual analysis scripts:

Figure	Generator	Status
pub_fig1_policy_comparison.png	regenerate_all_figures.py	✓ V2 data
pub_fig2_fleet_sensitivity.png	regenerate_all_figures.py	✓ V2 data
pub_fig3_demand_robustness.png	regenerate_all_figures.py	✓ V2 data
pub_fig4_service_sensitivity.png	regenerate_all_figures.py	✓ V2 data
pub_fig5_performance_heatmap.png	regenerate_all_figures.py	✓ V2 data
re-sponse_time_distribution_by_policy.png	regenerate_all_figures.py	✓ V2 data
fleet_sensitivity_dual.png	regenerate_all_figures.py	✓ V2 data
capacity_sensitivity_heatmap.png	generate_capacity_sensitivity_heatmap.py	⚠️ KNOWN BROKEN (data format issue)
cbd_equity_tradeoff_summary.png	regenerate_all_figures.py	✓ V2 data
cbd_robustness_enhanced.png	regenerate_all_figures.py	✓ V2 data
precinct_demand_heatmap.png	generate_precinct_demand_visualizations.py	✓ from processed data
precinct_demand_rates_improved.png	generate_precinct_demand_visualizations.py	✓ from processed data
project_summary_dashboard.png	generate_summary_dashboard.py	⚠️ V1 data refs, CAP=5
queue_comparison_by_policy.png	analyze_queue_metrics.py	⚠️ V1 data refs
queue_heatmap.png	analyze_queue_metrics.py	⚠️ V1 data refs
queue_vs_demand.png	analyze_queue_metrics.py	⚠️ V1 data refs

Figure	Generator	Status
<code>queue_vs_fleet_size.png</code>	<code>analyze_queue_metrics.py</code>	⚠ V1 data refs
<code>re-sponse_time_coverage_tradeoff.png</code>	<code>generate_tradeoff_improved.py</code>	⚠ CAP=5
<code>seasonal_decomposition.png</code>	<code>analyze_seasonal_patterns.py</code>	✅ from raw crash data
<code>seasonal_heatmap.png</code>	<code>analyze_seasonal_patterns.py</code>	✅ from raw crash data
<code>seasonal_patterns.png</code>	<code>analyze_seasonal_patterns.py</code>	✅ from raw crash data

Analysis Figures/Tables (`results/analysis/`)

Subdirectory	Content	Generator
<code>analysis/capacity_comparison/</code>	Full cap sweep (cap 1-5, K=20/40), 90+ CSV+PNG	<code>capacity_sensitivity_full_spectrum.py</code>
<code>analysis/cbd_focused_comparison/</code>	CBD vs Manhattan-wide comparison	<code>run_cbd_focused_optimization.py</code> ⚠ CAP=5
<code>analysis/distance_comparison/</code>	Haversine vs Manhattan distance	<code>run_distance_comparison_experiment.py</code> ⚠ CAP=5
<code>analysis/heatmaps/</code>	108 allocation heatmap PNGs + allocations	<code>generate_all_heatmaps.py</code>
<code>analysis/maps/</code>	K=40 allocation maps (P0/P1/P2)	<code>p0_spatial_analysis.py</code>
<code>analysis/simulation/production/</code>	V1 experiment CSVs (cap=5)	⚠ LEGACY — <code>run_production_experiments.py</code> (archived)
<code>analysis/simulation/cbd_experiment/</code>	CBD experiment results	<code>run_cbd_experiment.py</code>
<code>analysis/figures/</code>	~40 analysis figures	Various analysis scripts
<code>analysis/tables/</code>	~20 analysis tables	<code>analyze_production_results.py</code> ⚠ V1

=== 3. DOCS RECONCILIATION TARGET LIST ===

README.md

- Verify all paths point to `results/baseline/` as source of truth
- Confirm capacity=2, spatial P0 messaging

docs/core/ (30 files — check all, high-priority subset below)

Document	Priority	What to check
technical_report.md	● HIGH	All metric values, figure references, table references, K=20 baseline numbers
executive_summary.md	● HIGH	Headline metrics (RT=2.57min, coverage=99.6%) match regenerated data
executive_presentation.md	● HIGH	Same as executive summary
figure_trace_guide.md	● HIGH	All 40+ figure entries match regenerated filenames and generators
visualization_index.md	● MED	Cross-check with regenerated figure set
output_analysis.md	● MED	Metrics match baseline tables
experimental_design.md	● MED	Parameters match (cap=2, reps=30, K values)
verification_log.md	● MED	V&V results match regenerated JSONs
reproducibility_guide.md	● MED	Commands still work
decisions_log.md	● LOW	Historical, unlikely to need changes
assumptions_log.md	● LOW	Historical
conceptual_model.md	● LOW	Model description, unlikely to need changes
demand_model_spec.md	● LOW	Spec, not metrics
service_model_spec.md	● LOW	Spec, not metrics
optimization_formulation.md	● LOW	Check capacity=2 is documented
code_documentation.md	● LOW	API descriptions
testing_guide.md	● LOW	Test count may change
notebook_guide.md		Notebook descriptions

Document	Priority	What to check
	● LOW	
data_usage_guide.md	● LOW	Data paths
ARCHITECTURAL_MAP.md	● LOW	Structure description
DOCUMENTATION_INDEX.md	● LOW	Index of all docs
source_manifest.md	● LOW	File inventory
Other core docs	● LOW	Historical/reference

docs/analysis/ (14 files)

Document	Priority	What to check
capa-city_sensitivity_analysis.md	● MED	Metrics match regenerated capacity sweep
cbd_focused_optimization_analysis.md	● MED	Check if still references cap=5 data
cbd_robustness_analysis.md	● MED	Same
queue_analysis.md	● MED	Queue metrics match
optimization_results.md	● MED	Policy comparison metrics
policy_tradeoff_analysis.md	● MED	Tradeoff curve figures
re-search_questions_assessment.md	● MED	Answers reference correct metrics
dis-distance_metric_comparison.md	● LOW	Distance analysis
fleet_sensitivity_dual_investigation.md	● LOW	Fleet sensitivity discussion
fire-house_capacity_analysis.md	● LOW	Capacity discussion
eda_and_data_split_summary.md	● LOW	Data description
analysis_precinct_demand_trace.md	● LOW	Demand trace
cbd_comparison_and_validity_report.md	● LOW	CBD validity
alternative_analyses_summary.md	● LOW	Summary of alternatives

results/WHICH_FILES_TO_USE.md

- MED: Verify all paths and status markers are still correct after regeneration

=== 4. RISKS AND PREREQUISITES ===

Prerequisites

#	Prerequisite	How to Verify
P1	data/processed/ fully populated (13 files)	<code>ls data/processed/*.csv</code> <code>data/processed/*.geojson</code>
P2	Python environment with all deps	<code>pip install -r requirements.txt</code>
P3	PuLP solver available	<code>python -c "import pulp; pulp.LpSolverDefault"</code>
P4	SimPy installed	<code>python -c "import simpy"</code>
P5	~2GB RAM available for 810 simulation runs	<code>free -m</code>
P6	src/ems_readiness/ imports clean	<code>python -c "from ems_readiness.simulation.engine import EMSSimulation"</code>
P7	Git LFS files pulled (raw crash data not needed for simulation)	LFS only needed for data processing, not production runs

Risks

#	Risk	Impact	Mitigation
R1	V1/V2 data source mismatch — 7 analysis scripts still read V1 data (cap=5) from <code>results/analysis/simulation/production/</code>	⚠ HIGH: Analysis tables/figures won't match baseline	Fix scripts before regeneration to read from <code>results/baseline/simulation/all_results_raw.csv</code> , or explicitly mark as V1-legacy
R2	Broken capacity heatmap — <code>generate_capacity_sensitivity_heatmap.py</code> produces blank panels (known "Data format issue")	🟡 MED: One figure is known-broken	Debug the data format issue in the heatmap script before regeneration
R3	~20 possibly-broken figures — White fraction >92% in many baseline and top-level figures	🟡 MED: May indicate matplotlib rendering issues or sparse data	Visual inspection after regeneration; many are line/scatter plots with inherently white backgrounds
R4	Stochastic drift — New random seeds could produce slightly different metrics than what docs quote	🟢 LOW: Seeds are fixed (base=42), results should be deterministic	Verify key metrics (RT=2.57, coverage=99.6%) match within CI
R5	Long-running heatmap generation — <code>generate_all_heatmaps.py</code> generates 108+ maps	🟢 LOW: ~2 min	Run in background if needed
R6	Capacity sweep re-simulation — <code>capacity_sensitivity_full_spectrum.py</code> runs its own simulations	🟡 MED: Additional ~3 min, generates own data	Must run after baseline so comparative data is consistent
R7	CBD experiment uses dynamic parameters — <code>run_cbd_experiment.p</code>	🟢 LOW	Verify capacity=2 is used

#	Risk	Impact	Mitigation
	y may use different capacity defaults		

Validation Steps (Post-Regeneration)

#	Validation	Command/Check
V1	All 39 unit tests pass	<code>python -m pytest tests/ -v</code>
V2	Baseline has 810 simulation rows	<code>wc -l results/baseline/simulation/all_results_raw.csv</code> (should be 811 incl header)
V3	9 allocation files exist	<code>ls results/baseline/allocations/</code>
V4	4 verification JSONs populated	<pre>python -c "import json; [json.load(open(f'results/ baseline/simulation/veri- fication/0{i}_{n}.json')) for i,n in [(1,'toy_example'), (2,'zero_demand'), (3,'single_unit'), (4,'extreme_demand')]]"</pre>
V5	3 validation pilots populated	Check <code>results/baseline/simulation/validation_pilot/</code>
V6	Key metric check: K=20 P2 mean RT $\approx$ 2.57 min	<pre>python -c "import pandas as pd; df=pd.read_csv('results/ baseline/simulation/ all_results_raw.csv'); print(df[(df.K==20)&(df.pol- icy=='P2')].mean_response_t- ime.mean())"</pre>
V7	No broken figures (visual spot-check)	Open <code>results/baseline/figures/</code> <code>pub_fig1_policy_comparison.png</code> etc. in browser
V8	Capacity heatmap has actual data	Open <code>results/figures/capacity_sensitivity_heatmap.png</code> — should NOT show “Data format issue”

=== 5. RECOMMENDED BRANCH/PR STRATEGY ===

Strategy: Single Feature Branch with Staged Commits

```

main
└─ feature/canonical-regeneration
    └─ commit 1: "fix: redirect V1-referencing scripts to V2 baseline data"
        (fix 7 scripts that reference cap=5/V1 data)
    └─ commit 2: "fix: resolve capacity heatmap data format issue"
        (debug and fix generate_capacity_sensitivity_heatmap.py)
    └─ commit 3: "chore: regenerate baseline production outputs"
        (run_production_v2.py + verification + validation)
    └─ commit 4: "chore: regenerate analysis artifacts"
        (all analysis scripts in dependency order)
    └─ commit 5: "docs: reconcile documentation with regenerated outputs"
        (update all metrics, paths, figure references)
    └─ commit 6: "test: verify all tests pass with regenerated data"
  
```

Branch Naming

- feature/canonical-regeneration or chore/full-refresh-v2

PR Description Template

```

## Full Canonical Regeneration

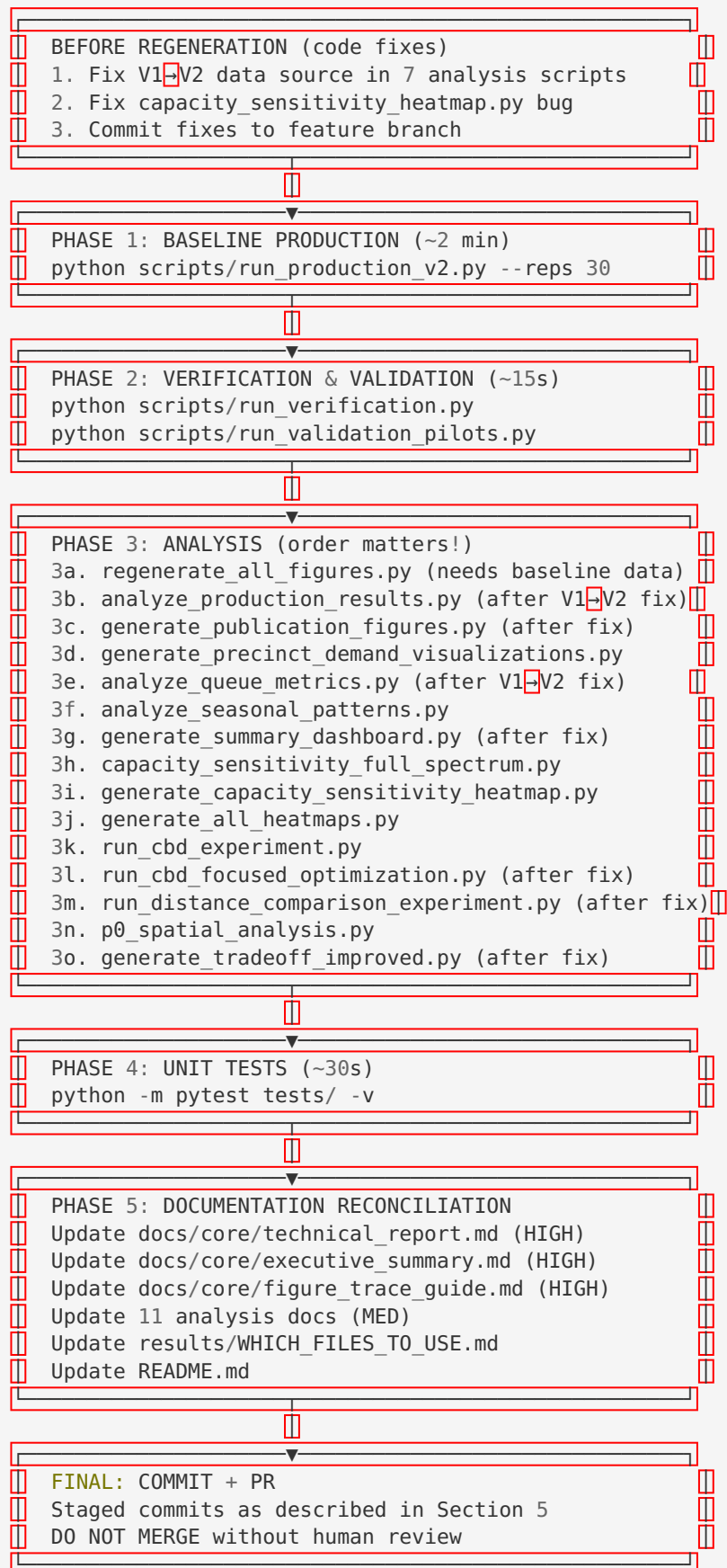
### What changed
- Fixed 7 analysis scripts to use V2 baseline data (cap=2) instead of V1 (cap=5)
- Fixed capacity sensitivity heatmap data format issue
- Regenerated all baseline simulation outputs (810 runs)
- Regenerated all analysis figures and tables
- Reconciled all documentation with fresh outputs

### Verification
- [ ] 39 unit tests pass
- [ ] 810 simulation rows in all_results_raw.csv
- [ ] Key metrics match: K=20 P2 RT ≈ 2.57 min, coverage ≈ 99.6%
- [ ] Capacity heatmap renders correctly
- [ ] All 3 validation pilots pass directional checks
- [ ] Visual spot-check of 5 publication figures
  
```

Important Notes

- **Do NOT force-push** to main
- **Do NOT auto-merge** — requires human review
- Keep results/archive/ untouched (historical reference)
- The results/analysis/simulation/production/ V1 data can be preserved in results/archive/ if needed for historical comparison

=== 6. EXECUTION ORDER SUMMARY ===



=== 7. SCRIPTS REQUIRING V1→V2 DATA SOURCE FIX ===

These 7 scripts currently reference `results/analysis/simulation/production/` (V1 cap=5 data) and need to be redirected to `results/baseline/simulation/all_results_raw.csv` (V2 cap=2 data):

#	Script	Current Data Source	Fix Required
1	<code>ana-lyze_production_results.py</code>	<code>results/analysis/simulation/production/exp{1..4}*.csv</code>	Redirect to V2 or refactor to read <code>all_results_raw.csv</code>
2	<code>generate_publication_figures.py</code>	Same V1 path	Same fix
3	<code>ana-lyze_queue_metrics.py</code>	V1 experiment CSVs	Same fix
4	<code>generate_summary_dashboard.py</code>	V1 path + CAP=5	Same fix + update CAPACITY constant
5	<code>run_cbd_focused_optimization.py</code>	Uses CAP=5 internally	Change to CAP=2
6	<code>run_distance_comparison_experiment.py</code>	Uses CAP=5 internally	Change to CAP=2
7	<code>generate_tradeoff_improved.py</code>	Uses CAP=5 internally	Change to CAP=2

Estimated fix effort: ~1 hour of careful script editing + testing

This document is a planning artifact. No regeneration has been executed.